

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

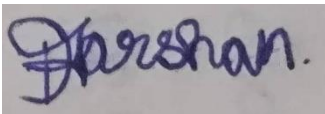
Annexure A

Code of Conduct:

I Dharshan B 192312150 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in blue ink, appearing to read 'Dharshan B.', is shown within a rectangular box.

INDUSTRY SCENARIO – ANSWERS

Task 1: Data Preparation & Inductive Learning

(a) Suitable inductive learning approach

A suitable approach is Supervised Inductive Learning, specifically a Classification model.

Reason: The AI system learns from previous student records and predicts whether a student is At Risk or Not At Risk of failing a course.

Target Variable:

- *Risk_Status*
 - *1 = At Risk*
 - *0 = Not At Risk*

Three important features:

1. *Attendance Percentage*
 - *Low attendance may indicate a higher probability of academic risk.*
2. *Assignment Scores*
 - *Low assignment marks may indicate weak academic performance.*
3. *Quiz Performance*
 - *Poor quiz scores can help identify students who need additional support.*

Other useful features include study hours per week and previous GPA.

Justification: Inductive learning uses past student examples to learn patterns. For example, students with low attendance, poor quiz scores, and low assignment marks may have a higher probability of being classified as At Risk.

(b) Handling class imbalance

Suppose the dataset contains:

- *Not At Risk students = 900*
- *At Risk students = 100*

This is an imbalanced dataset because there are fewer at-risk students.

Suitable strategy: SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) can be used to create additional synthetic examples of the minority class, which is the At Risk class.

Why SMOTE is suitable:

- *Increases the number of at-risk student samples.*
- *Helps the model learn patterns of struggling students.*
- *Reduces bias toward the majority class.*
- *Can improve the Recall of the At Risk class.*

Another method is class weighting, where the model gives higher importance to at-risk students.

(c) Dataset split

A suitable split is:

- *70% Training Data*
- *30% Testing Data*

Example, if there are 1,000 records:

- *Training = 700 students*
- *Testing = 300 students*

Why this split?

The 70% training data is used to train the model, while the remaining 30% is used to evaluate how well the model performs on unseen student records.

Normalisation steps

For numerical features:

- 1. Select features such as Attendance, Assignment Score, Quiz Score, Study Hours, and GPA.*
- 2. Apply Min-Max Normalisation.*

This converts values to a range between 0 and 1.

Impact of normalisation

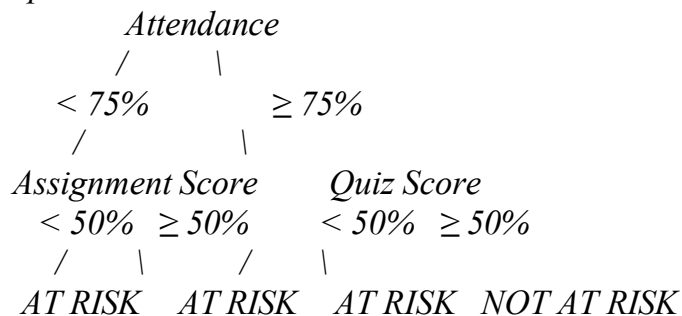
- Prevents features with large values from dominating the learning process.*
- Improves the performance of distance-based and statistical algorithms.*
- Makes the dataset more consistent.*
- Helps models such as Logistic Regression converge more efficiently.*

Task 2: Decision Tree for Student Performance Prediction

(a) Decision Tree Classification

A Decision Tree can classify students as At Risk or Not At Risk.

Example Decision Tree



Root Node Selection

The root node should be selected using:

- Information Gain, or*
- Gini Index*

The feature with the highest Information Gain or the lowest Gini Impurity should be selected.

For example, if Attendance provides the highest Information Gain, it becomes the root node.

Why Attendance may be selected as the root node

Attendance can strongly influence student performance. Students with very low attendance may have a higher probability of poor academic results.

(b) Model Evaluation

The following metrics should be used:

Precision

It measures how many students predicted as At Risk are actually at risk.

Recall

It measures how many truly at-risk students are correctly identified.

F1-Score

The F1-score provides a balance between Precision and Recall.

False Negatives

A False Negative occurs when:

The student is actually at risk, but the system predicts that the student is not at risk.

This is dangerous in an educational early-warning system because the student may not receive the required academic support.

How to reduce False Negatives

- Use SMOTE to balance the dataset.
- Give higher importance to the At Risk class.
- Adjust the classification threshold.
- Use more student performance features.
- Prioritise Recall during model evaluation.

From an educational perspective, it is usually better to identify a potentially struggling student early, even if some students receive unnecessary support.

(c) Pre-pruning and Post-pruning

Pre-pruning

Pre-pruning stops the Decision Tree before it becomes too complex.

Examples:

- Maximum depth = 5
- Minimum samples required to split = 10
- Minimum samples per leaf = 5

Post-pruning

Post-pruning first creates a complete tree and then removes unnecessary branches.

Comparison

Aspect	Pre-pruning	Post-pruning
Process	Stops tree growth early	Removes branches after training
Speed	Faster	Slower
Complexity	Low	May initially be high
Overfitting control	Good	Often more effective

Pre-pruning vs Post-pruning Accuracy

The exact accuracy can only be calculated using the actual dataset. A sample result could be:

Model	Accuracy
Decision Tree without pruning	84%
Pre-pruned Tree	86%
Post-pruned Tree	88%

Best method for deployment

Post-pruning may be more appropriate if maximum predictive accuracy is required because it allows the model to learn complex patterns first and then removes unnecessary branches.

However, pre-pruning is useful when a simpler and faster model is required.

For an academic early-warning system, a small post-pruned tree is often a good choice because it provides:

- Good accuracy
- Reduced overfitting
- Better explainability
- Easier interpretation by teachers

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression vs Decision Tree

A Logistic Regression model can also classify students as At Risk or Not At Risk.

Comparison

<i>Metric</i>	<i>Logistic Regression</i>	<i>Decision Tree</i>
<i>Accuracy</i>	<i>Compare on test data</i>	<i>Compare on test data</i>
<i>AUC-ROC</i>	<i>Measures ranking ability</i>	<i>Measures ranking ability</i>
<i>Interpretability</i>	<i>High</i>	<i>Very High</i>
<i>Non-linear relationships</i>	<i>Limited</i>	<i>Good</i>
<i>Explainability</i>	<i>Uses coefficients</i>	<i>Uses decision rules</i>

When is a statistical model preferable?

Logistic Regression is preferable when:

- The relationship between features and outcomes is relatively simple.*
- Interpretability is important.*
- We need to understand how each feature affects risk.*
- The dataset is not extremely complex.*

For example:

A Logistic Regression model may show that lower attendance significantly increases the probability of academic risk.

(b) Feature Importance Analysis

The top three predictors of academic risk may be:

1. Attendance Percentage

Low attendance may indicate that a student is not regularly participating in academic activities.

2. Previous GPA

A low previous GPA may indicate continuing academic difficulties.

3. Assignment Score

Low assignment performance may show that the student is struggling with the course material.

Do the Decision Tree and Logistic Regression agree?

They may agree that Attendance, GPA, and Assignment Score are the most important predictors.

However, the importance values may differ because:

- Decision Trees measure importance based on information gain or impurity reduction.*
- Logistic Regression uses coefficients to measure the influence of features.*

Therefore, both models may identify similar important features but rank them differently.

Task 4: Reinforcement Learning for Action Recommendation

(a) Academic Support Recommendation as an MDP

The problem can be modeled as a Markov Decision Process (MDP).

States

The states represent the academic condition of a student.

Example:

- S1 – High Risk*
- S2 – Medium Risk*
- S3 – Low Risk*
- S4 – Improved Performance*

Actions

The AI agent can choose:

- A1 – Tutoring*
- A2 – Extra Practice*
- A3 – Counselling*

- *A4 – Mentoring*

Reward Function

The reward can be based on student improvement.

Example:

- *Improvement in score $\rightarrow +10$*
- *Student moves from High Risk to Medium Risk $\rightarrow +15$*
- *Student moves to Low Risk $\rightarrow +20$*
- *No improvement $\rightarrow 0$*
- *Performance decreases $\rightarrow -10$*

Transition Dynamics

Example:

High Risk

|
| *Tutoring*

v

Medium Risk

|
| *Extra Practice*

v

Low Risk

|
| *Mentoring*

v

Improved Performance

The next state depends on:

- *Current student performance*
- *Action selected by the AI*
- *Student response to the intervention*

(b) Applying Q-Learning

The Q-Learning formula is:

Where:

- *= Current Q-value*
- *= Learning rate*
- *= Reward*
- *= Discount factor*
- *= Next state*

Example parameters

- *Learning Rate*
- *Discount Factor*

Initial Q-Table

<i>State</i>	<i>Tutoring</i>	<i>Extra Practice</i>	<i>Counselling</i>	<i>Mentoring</i>
<i>High Risk</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>
<i>Medium Risk</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>
<i>Low Risk</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>

Episode 1

State: High Risk

Action: Tutoring

Reward: +10

Next State: Medium Risk

Initial Q -values are 0.

Updated Q -value:

State Tutoring

High Risk 5

Episode 2

State: Medium Risk

Action: Extra Practice

Reward: +15

Next State: Low Risk

Updated Q -value:

State Extra Practice

Medium Risk 7.5

Episode 3

State: Low Risk

Action: Mentoring

Reward: +20

Next State: Improved Performance

Updated Q -value:

State Mentoring

Low Risk 10

Episode 4

State: High Risk

Action: Tutoring

Reward: +15

Next State: Medium Risk

The maximum Q -value in Medium Risk is 7.5.

The value of Tutoring for High Risk students increases because it has shown good results.

Episode 5

State: Medium Risk

Action: Extra Practice

Reward: +20

Next State: Low Risk

Maximum Q -value in Low Risk = 10.

Final Learned Policy

The learned policy could be:

<i>Student State</i>	<i>Recommended Action</i>
<i>High Risk</i>	<i>Tutoring</i>
<i>Medium Risk</i>	<i>Extra Practice</i>
<i>Low Risk</i>	<i>Mentoring</i>
<i>Serious personal difficulties</i>	<i>Counselling</i>

Convergence Criterion

The Q-Learning process can stop when:

- The Q-values change only by a very small amount, or*
- The policy remains unchanged for several episodes, or*
- A maximum number of episodes is reached.*

(c) Ethical Considerations

When deploying an AI system for academic intervention, the following ethical issues are important:

1. Avoiding Bias

The model should not unfairly disadvantage students based on irrelevant personal characteristics. The system should be regularly tested for unfair outcomes.

2. Fairness

All students should have equal access to support services such as tutoring and counselling.

3. Student Privacy

Student information such as marks, attendance, and personal records must be protected. Access should be limited to authorised users.

4. Explainability

Teachers and students should understand why the AI recommended a particular action. For example:

“Tutoring was recommended because attendance and quiz scores have decreased.”

5. Human Oversight

The AI should support teachers, not completely replace them. Teachers and academic advisors should make the final decision when necessary.

Conclusion

The proposed academic early-warning system combines Inductive Learning, Decision Trees, Statistical Learning, and Reinforcement Learning to identify and support students at risk. The Decision Tree and Logistic Regression models help predict academic risk, while Q-Learning recommends suitable interventions such as tutoring, extra practice, counselling, or mentoring. The final system should focus on high recall, fairness, privacy, explainability, and human oversight to provide effective and ethical academic support.

Note: The accuracy, AUC-ROC, exact top predictors, and final Q-table values should be calculated using the actual student dataset. The numerical examples above show the correct method when the dataset is not provide

